

# NORTHEAST INDIA LOGISTICS RESILIENCE PLATFORM (NER-LRP)

## AI & GIS-Driven Regional Logistics Control Tower for Northeast India

Real-Time Ingestion | Geospatial ML Risk Engine | Cargo Prioritization | Offline PWA | Multilingual Broadcast

### 1. Executive Overview & Problem Context

The Northeast Region (NER) of India is characterized by extremely steep mountainous topography (>70% hilly), heavy monsoon precipitation, frequent flash floods, landslides, and single-artery national highways (e.g., NH-10 into Sikkim, NH-6 through Meghalaya into Mizoram/Tripura, NH-29 into Nagaland). Disruptions frequently choke essential supply chains, inflating local commodity costs by **over 20%** and delaying life-saving pharmaceuticals and medical oxygen to remote district hospitals.

**The Solution:** NER-LRP establishes an end-to-end **Sense-Predict-Decide-Act** Control Tower combining live meteorological feeds (Open-Meteo & IMD APIs), NASA SRTM slope profiles, OpenStreetMap road topology, and Geological Survey of India (GSI) disaster records to compute explainable hazard scores, prioritize critical medical cargo, enable offline-first field incident reporting, and broadcast multilingual voice alerts across 6 regional languages.

### 2. Data Authenticity & Architecture Matrix

| Layer / Component  | Data Provider / Technology        | Authenticity / Real-World Basis                                              |
|--------------------|-----------------------------------|------------------------------------------------------------------------------|
| Live Weather Feeds | Open-Meteo & IMD APIs             | 100% Real-Time (Rainfall mm, 48h cumulative rain, wind speed, soil moisture) |
| Mountain Slopes    | NASA SRTM Digital Elevation Model | Real Slope Angles (e.g. 38.5° NH-10 Teesta, 36.5° NH-6 Sonapur Pass)         |
| Highway Topologies | OpenStreetMap (OSM) Graph Engine  | Real Road Geometries across 8 Northeast States (4,820 KM monitored)          |
| Disaster Zones     | Geological Survey of India (GSI)  | Verified Historical Landslide Hotspots (Sonapur, 29th Mile, Chumukedima)     |
| AI Risk Scoring    | Scikit-Learn Gradient Boosting ML | Trained ML Model with explainable factor contributions (0-100 score)         |
| Offline Field Sync | IndexedDB & LocalStorage PWA      | Offline-First Local Queue with idempotent batch sync upon reconnect          |
| Voice Broadcaster  | Web Speech Synthesis API          | Live audible speech synthesis in 6 regional Northeast languages              |

## 3. Deep Dive into Every Feature & How to Use It

### Feature 1: Interactive GIS Command Tower Map

The central dashboard renders a high-performance Leaflet dark map of Northeast India with color-coded highway lines, weather radar precipitation overlays, and animated moving fleet markers.

- **Basemap Style Selector:** Located in the top-right toolbar. Switch between *ESRI Dark Canvas*, *OpenStreetMap Detailed*, and *ESRI Satellite Terrain* (100% free with zero watermarks).
- **Color Status Coding:** Green (Safe, <35% risk), Yellow (Watch, 36-60%), Orange (Warning, 61-80%), Red (Blocked, >80% risk / active landslide), and Dashed Emerald (Dynamic AI Alternate Bypass).
- **Interactive Inspection:** Click on any highway line to open the AI Risk Explainability drawer, or click any truck icon to inspect CAN-bus telemetry, speed, and cold-chain temperature.

### Feature 2: 1-Click Disruption Simulator

Demonstrates real-time system reaction to sudden disasters without having to wait for a real-world landslide during evaluation.

- **Scenario 1 (NH-6 Sonapur Landslide):** Triggers 3,800m<sup>3</sup> hill collapse -> NH-6 turns RED -> System intercepts Medical Oxygen Convoy NER-MED-8910 and diverts via NH-27 Haflong bypass (+55m ETA, +76% safety factor).
- **Scenario 2 (NH-10 Kalijhora Flash Flood):** Teesta overtopping blocks Sikkim highway -> NDRF relief diverted via Lava-Gorubathan ridge (+90m ETA).
- **Scenario 3 (NH-29 Chumukedima Mudslide):** Mudflow restricts Nagaland lifeline -> Food grain fleet diverted via Bokajan-Wokha ridge (+105m ETA).
- **Scenario 4 (48h Monsoon Inundation):** Simulates regional 180-250mm downpour escalating hazard scores across all 7 state gateways.
- **Reset Button:** Instantly normalizes all corridors back to live meteorological monitoring baseline.

### Feature 3: Cargo Fleet Prioritization & Telemetry

Unlike standard GPS apps that treat all vehicles equally, NER-LRP enforces multi-tier cargo criticality weights:

- **Priority 1 (Critical Medical):** 85% Safety Weight / 15% Travel Time. Prioritizes life-support supplies, taking detours to bypass high-risk zones.
- **Priority 2 (Disaster Relief):** 60% Safety Weight / 40% Travel Time. Balanced optimization for rations and water purification units.
- **Priority 3 (Commercial Bulk):** 25% Safety Weight / 75% Travel Time. Optimizes for lowest transit time unless road is physically blocked.

## Feature 4: Offline-First Field Officer PWA

Engineered for patrol officers and Border Roads Organisation (BRO) personnel in remote valleys with zero cellular reception:

- **Offline Queue:** When offline, reports (photo, GPS coordinate lock, severity rating 1-10) save locally to IndexedDB with **Sync Pending** status.
- **Automatic Batch Sync:** Upon entering cellular range, the app automatically batch-syncs to the FastAPI backend, transitions status to **Synced**, and drops a pulsating incident marker on the central Command Tower map.

## Feature 5: Multilingual Alert Hub & Voice Broadcaster

Translates emergency road advisories into 6 Northeast regional languages: **English, Hindi, Assamese (ꯀꯪꯂꯩꯃꯥ), Mizo (Mizo ꯁꯤꯟꯪ), Manipuri (ꯀꯪꯂꯩꯃꯥ), and Bengali (ꯀꯪꯂꯩꯃꯥ).**

- **Audible Speech Synthesis:** Click **■ Listen Audio** to hear the warning spoken aloud in the selected regional language.
- **Driver Dispatch Payloads:** Generates pre-formatted WhatsApp and GSM-7 SMS messages ready to dispatch to en-route drivers.

## Feature 6: Explainable AI (XAI) Risk Factor Drawer

Eliminates 'black box' AI by providing mathematical factor attribution for every hazard score: e.g. 96/100 Risk = 48h Rain (310mm) [42%] + Slope (36.5°) [32%] + GSI Hotspot [18%] + Active Reports [8%].

#### 4. Step-by-Step 3-Minute Hackathon Demo Script

**[0:00 - 0:30] Hook:** 'Over 70% of Northeast India is mountainous with single-artery lifelines. When landslides strike NH-6 or NH-10, life-saving medicines are blocked and local food prices jump 20%. Today we present NER-LRP—an AI and GIS-driven Control Tower.'

**[0:30 - 1:15] GIS Tower:** 'Our system connects live to Open-Meteo weather APIs and NASA SRTM slope profiles across 4,820 km of Northeast corridors. Every highway on this dark GIS map is dynamically color-coded by our Gradient Boosting ML model.'

**[1:15 - 2:00] Simulation:** 'Let's trigger a landslide at NH-6 Sonapur Tunnel. [Click button]. NH-6 turns RED. Unlike Google Maps, our system intercepts Convoy NER-MED-8910 carrying Emergency Oxygen and reroutes it via the NH-27 Haflong bypass (+55m delay).'

**[2:00 - 2:35] Offline PWA:** 'In zero-internet valleys, our Offline PWA allows patrol officers to submit photo damage reports. [Toggle Offline ON -> Submit -> Toggle Offline OFF]. Notice how it auto-syncs the moment connectivity is restored.'

**[2:35 - 3:00] Multilingual:** 'Finally, we broadcast advisories in 6 regional languages [[Click Assamese -> Listen Audio](#)] so drivers receive voice alerts hands-free. Thank you!'

## 5. Judge Q&A; Stress-Test Defense

### Q: Why can't authorities just use Google Maps?

**A:** Google Maps is a consumer navigation app; it does not predict landslides from rainfall/slope data before they occur, does not prioritize critical medicine over cement, and lacks offline field reporting and regional multilingual audio broadcasting.

**Q: What data is real vs simulated?**

**A:** Weather feeds (Open-Meteo API), mountain slopes (SRTM DEM), road geometry (OpenStreetMap), and landslide history (GSI) are 100% real. As stated in the Problem Statement, truck telematics and test disaster triggers are simulated for live evaluation demonstrations.

**Q: Why use Machine Learning instead of IF/ELSE rules?**

**A:** Landslides depend on non-linear relationships between 48h cumulative rain, rain intensity, slope angle, and soil saturation. Gradient Boosting learns these multi-factor patterns and provides explainable factor weightings for every highway.